

Mathematical Foundations of Linear Regression

Contents

1	Introduction	2
2	Problem Statement	2
3	Matrix Representation	2
4	Linear Model	3
5	Cost Function	3
6	Gradient	3
7	Why the Transpose?	3
8	Gradient Descent	4
9	Dimension Summary	4
10	Conclusion	4

1 Introduction

Linear Regression is one of the most fundamental algorithms in Machine Learning. Its goal is to model the relationship between an input variable and an output variable using a linear function.

2 Problem Statement

Suppose we have a dataset composed of m observations:

$$(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m) \quad (1)$$

We seek a line of the form

$$\hat{y} = ax + b \quad (2)$$

that best fits the observed data.

3 Matrix Representation

The input vector is

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}. \quad (3)$$

Similarly,

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}. \quad (4)$$

The design matrix is

$$X = \begin{pmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \vdots \\ x_m & 1 \end{pmatrix}. \quad (5)$$

The column of ones allows us to include the intercept term b . Without this column, the model would become

$$\hat{y} = ax \quad (6)$$

and the regression line would be forced to pass through the origin. The parameter vector is

$$\theta = \begin{pmatrix} a \\ b \end{pmatrix}. \quad (7)$$

4 Linear Model

The model can be written as

$$\hat{y} = X\theta. \quad (8)$$

Dimension check:

$$(m \times 2)(2 \times 1) = (m \times 1). \quad (9)$$

Thus, the prediction vector has the same dimension as y .

5 Cost Function

The residual error is

$$e = \hat{y} - y. \quad (10)$$

The Mean Squared Error cost function is

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2. \quad (11)$$

In matrix form:

$$J(\theta) = \frac{1}{2m} (X\theta - y)^T (X\theta - y). \quad (12)$$

6 Gradient

The gradient of the cost function is

$$\nabla J(\theta) = \frac{1}{m} X^T (X\theta - y). \quad (13)$$

The gradient indicates the direction of the steepest increase of the cost function.

7 Why the Transpose?

The error vector has dimension

$$X\theta - y \in \mathbb{R}^{m \times 1}. \quad (14)$$

The parameter vector has dimension

$$\theta \in \mathbb{R}^{2 \times 1}. \quad (15)$$

Since

$$X^T \in \mathbb{R}^{2 \times m}, \quad (16)$$

we obtain

$$X^T (X\theta - y) = (2 \times m)(m \times 1) = (2 \times 1). \quad (17)$$

Therefore, the transpose ensures that the gradient has the correct dimensions.

8 Gradient Descent

The parameters are updated according to

$$\theta \leftarrow \theta - \alpha \nabla J(\theta), \quad (18)$$

where α is the learning rate.

9 Dimension Summary

Object	Dimension
x	$(m, 1)$
y	$(m, 1)$
X	$(m, 2)$
θ	$(2, 1)$
$X\theta$	$(m, 1)$
$X\theta - y$	$(m, 1)$
X^T	$(2, m)$
$X^T(X\theta - y)$	$(2, 1)$

10 Conclusion

Linear Regression combines linear algebra, calculus, and optimization. Understanding the matrix formulation, the cost function, and gradient descent provides a solid foundation for more advanced Machine Learning algorithms.